

Code Reference: README.md

A source-level explanation with function details, file communication, variables, endpoints, and debugging notes.

Item	Explanation
File	README.md
Purpose	Main README for the standalone builder application: install, updates, workspaces, text editing, publishing, build commands, branch model, and uninstall.
Lines	238
Size	22,595 bytes
Talks to	Cloudflare/AI layer, GitHub/release layer
Functions	0
Variables/constants	0
API mentions	0
Signals	67

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Cloudflare or AI at line 17	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: It explains the high-level design, generated block diagrams, software-engineering decisions, shell commands, programming syntax, how the important files communicate, what each build tool owns, GitHub workflows, Electron,
Installer at line 21	Windows setup, update, shortcut, or installation behavior. Evidence: Download the latest Windows installer or portable executable from:
Network request at line 24	Frontend-backend communication or public web/API calls. Evidence: https://github.com/otienomaurice/omb-portfolio-builder/releases/latest
Installer at line 29	Windows setup, update, shortcut, or installation behavior. Evidence: - `OMB-Portfolio-Builder-Setup-latest-x64.exe` for the Windows installer.
Installer at line 32	Windows setup, update, shortcut, or installation behavior. Evidence: The normal installer default is a per-user AppData location:
Installer at line 35	Windows setup, update, shortcut, or installation behavior. Evidence: C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder
Installer at line 41	Windows setup, update, shortcut, or installation behavior. Evidence: C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder\OMB Portfolio Builder.exe
Installer at line 44	Windows setup, update, shortcut, or installation behavior. Evidence: The installer uses a Windows wizard with Back, Next, Cancel, progress, and Finish. It can create Desktop and Start Menu shortcuts, and it writes a normal Windows uninstall entry. The default per-user install does not wri
Git command at line 46	Repository state, publishing, branch checks, or release automation. Evidence: Installed users do not need Node.js or pnpm. The packaged desktop app includes its runtime. Git for Windows with Git Credential Manager is needed only for publishing to a Git repository. If Git or Git Credential Manager
Installer at line 50	Windows setup, update, shortcut, or installation behavior. Evidence: The app checks GitHub Releases for newer builder versions. When a newer release exists, the app shows an update dialog. Choosing **Update** downloads the installer, closes the running builder, updates the existing instal
Network request at line 57	Frontend-backend communication or public web/API calls. Evidence: https://github.com/otienomaurice/omb-portfolio-builder/releases
Installer at line 62	Windows setup, update, shortcut, or installation behavior. Evidence: On Windows, the app keeps application files and editable content under AppData:
Installer at line 64	Windows setup, update, shortcut, or installation behavior. Evidence: - `C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder` stores the installed desktop application.

Item	Explanation
Installer at line 65	Windows setup, update, shortcut, or installation behavior. Evidence: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\builder` stores the builder workspace, local drafts, uploaded files, templates, previews, and editing tools.
Git command at line 66	Repository state, publishing, branch checks, or release automation. Evidence: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\portfolio` stores the publish mirror used by Git-based publishing.
Installer at line 67	Windows setup, update, shortcut, or installation behavior. Evidence: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\compile-code` stores project compile-workspace source files and run folders created by the Compile Code tool.
Installer at line 69	Windows setup, update, shortcut, or installation behavior. Evidence: The installer and updater treat the AppData builder and portfolio folders as managed workspaces, not duplicate app installs.
Security or auth at line 71	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: The builder works offline for editing, project creation, file attachment, local previews, and draft saves. Publishing requires network access and repository write permission.
Rich text at line 86	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Project and portfolio-area overview editors support rich text editing with selection-level formatting, links, images, formulas, syntax-highlighted code blocks, hard line breaks, copy, paste, cut, and select all.
Rich text at line 88	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Front-page, profile, contact, fun facts, and project overview fields use the same selection-level text behavior: highlight the exact words to edit, then use the right-click menu or floating selection inspector to change
Compiler or simulator at line 90	Part of the Compile Code workspace or language/tool detection. Evidence: Code support is available from the rich editor right-click menu. Use Code block or Paste as code for C, C++, SystemVerilog, LTspice, Java, JavaScript, Python, and HTML. Code blocks store the selected language, pr
Compiler or simulator at line 92	Part of the Compile Code workspace or language/tool detection. Evidence: Projects also include a separate Compile Code workspace. This is local builder tooling, not a portfolio text block. Each project behaves like a small IDE workspace: the left explorer lists the source files, the main
Compiler or simulator at line 94	Part of the Compile Code workspace or language/tool detection. Evidence: The Compile Code workspace has separate Compile , Build project , and Run controls. Verilog/SystemVerilog also show Simulate and Show scope . Compile checks or builds the active source. Build project use
Compiler or simulator at line 98	Part of the Compile Code workspace or language/tool detection. Evidence: Verilog and SystemVerilog simulation requires a testbench. Mark HDL files as Design or Testbench , or use Add testbench to create a starter bench with <code>\$dumpfile`</code> and <code>\$dumpvars`</code> . HDL Compile can syntax-check/
Compiler or simulator at line 106	Part of the Compile Code workspace or language/tool detection. Evidence: For example, a SystemVerilog file attached to a design section might be saved as <code>docs\laser-link\design\filter.sv`</code> . The upload status in the builder shows the generated relative path after a file is saved.
Rich text at line 112	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Dark mode applies to builder panels, dialogs, guide windows, menus, status chips, and window/icon controls. Editing fields and selected project appearances keep their own colors so typed content, field formatting, and pr
Git command at line 116	Repository state, publishing, branch checks, or release automation. Evidence: Publishing is controlled by Git repository write access. The builder does not embed owner passwords or private credentials.
Git command at line 124	Repository state, publishing, branch checks, or release automation. Evidence: https://github.com/YOUR-USERNAME/YOUR-USERNAME.github.io.git
Security or auth at line 127	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: 3. Click Authenticate target .
Git command at line 128	Repository state, publishing, branch checks, or release automation. Evidence: 4. Complete GitHub/Git Credential Manager sign-in if prompted.
Security or auth at line 132	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Successful authorization is cached locally per repository and branch for about one day. If the same target receives more than three successful authorizations in one week, the local trust window can extend to 30 days for
Git command at line 136	Repository state, publishing, branch checks, or release automation. Evidence: Install Git for Windows, Node.js, and npm. Then run:
Git command at line 139	Repository state, publishing, branch checks, or release automation. Evidence: git clone https://github.com/otienomaurice/omb-portfolio-builder.git
Installer at line 156	Windows setup, update, shortcut, or installation behavior. Evidence: npm run installer
Installer at line 168	Windows setup, update, shortcut, or installation behavior. Evidence: When <code>`main`</code> receives a release-ready version, GitHub Actions builds the Windows installer and portable executable, creates or updates the <code>`builder-v<version>`</code> GitHub Release, and uploads stable <code>`latest`</code> asset names.
Installer at line 186	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`main`</code> Consumer/release branch for finished installer and public app code. Initial repository branch None Release-ready only

Item	Explanation
Rich text at line 192	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>`codex/manual-update-categories-preview`</code> Adds manual update checks, custom project categories, link-safe rich paste handling, project-only website previews, broad duplicate-install detection, and a main release workflow
Security or auth at line 193	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>`codex/fix-publish-target-auth`</code> Makes publishing target setup authenticate before saving, resolves the target default branch before access checks, and blocks known local builder workspaces during installation. <code>`dev`</code>
Security or auth at line 194	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>`codex/cache-daily-auth-and-restore-profile`</code> Adds daily and extended publishing authorization caching, clarifies publishing target fields, keeps active-machine profile fields populated, and enforces development-only
Installer at line 195	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`codex/automatic-installer-update-flow`</code> Makes the update popup run the installer automatically, strengthens existing-install detection, and publishes stable latest installer asset names for downloads. <code>`development`</code>
Installer at line 196	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`codex/split-builder-portfolio-workspaces`</code> Separates local builder files from publish-mirror files, syncs publish-safe assets, and bumps the next installer release version. <code>`development`</code> <code>`development`</code> Active
Installer at line 197	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`codex/guard-latest-builder-download`</code> Keeps the stable latest installer asset URL aligned with release assets. <code>`development`</code> <code>`development`</code> Active
Security or auth at line 198	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>`codex/secure-builder-editor-shell`</code> Adds native desktop menus, light app chrome, scoped publishing authorization cache, and smoother rich-editor copy, paste, cut, and image placement behavior. <code>`development`</code> <code>`dev`</code>
Installer at line 200	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`codex/local-desktop-shortcut`</code> Forces installer-created shortcuts onto the physical Windows desktop instead of redirected OneDrive Desktop folders. <code>`development`</code> <code>`development`</code> Active
Installer at line 204	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`codex/fix-update-auto-restart`</code> Makes the app update flow quit the desktop app cleanly, run the installer, and relaunch the updated executable automatically. <code>`development`</code> <code>`development`</code> Merged into <code>`development`</code>
Installer at line 205	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`codex/fix-update-launch-handoff`</code> Launches the updater through a detached Windows command handoff so the installer/relaunch script survives app shutdown. <code>`development`</code> <code>`development`</code> Merged into <code>`development`</code>
Rich text at line 208	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>`codex/fix-editor-installer-update-flow`</code> Fixes rich-editor context menus, prevents pasted links from rendering as formulas, adds safe fullscreen exit behavior, and lets installers update in place when old uninstaller
Installer at line 212	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`codex/skip-old-uninstaller-on-update`</code> Prevents in-app silent updates from launching stale uninstallers and adds an updater timeout for hung installers. <code>`development`</code> <code>`development`</code> Active
Security or auth at line 214	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>`codex/security-and-context-menu-reporting`</code> Improves text-field context menus, adds builder security/download reporting, and prepares publishable website security headers. <code>`development`</code> <code>`development`</code> Active
Rich text at line 215	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>`codex/uniform-rich-text-controls`</code> Makes right-click text formatting consistent for front-page, profile, and contact fields, persists field styles into previews and published portfolios, and expands common color choices
Security or auth at line 216	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>`codex/code-block-preferences`</code> Adds code-block authoring with syntax highlighting, source/basic paste modes, parser rendering, and builder Preferences for light/dark mode. <code>`codex/rich-profile-frontpage-editors`</code> <code>`dev`</code>
Rich text at line 217	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>`codex/full-builder-dark-mode`</code> Expands builder dark mode across panels, dialogs, menus, status chips, and icon controls while preserving input-field colors and project appearance templates. <code>`codex/code-block-prefer`</code>
Cloudflare or AI at line 218	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>`codex/per-user-update-migration`</code> Moves the installer/update model to the current user's app location and prefers per-user install records over stale machine-wide records. <code>`codex/full-builder-dark-mode`</code> <code>`development`</code>
Installer at line 219	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`codex/appdata-workspace-root`</code> Standardizes future installs and managed builder/portfolio workspaces under AppData and migrates legacy OMB app-folder installs forward. <code>`codex/per-user-update-migration`</code> <code>`development`</code>
Installer at line 220	Windows setup, update, shortcut, or installation behavior. Evidence: <code>`codex/project-code-compile-workspace`</code> Adds a project-local Compile Code workspace with source import, beautify, compiler detection, save, and terminal-style compile/run output. <code>`codex/appdata-workspace-root`</code> <code>`dev`</code>
Cloudflare or AI at line 221	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>`codex/fix-public-ai-worker-endpoint`</code> Adds the missing Cloudflare Worker backend files and wires the generated website to <code>`/api/portfolio-ai`</code> . <code>`development`</code> <code>`development`</code> Active

Item	Explanation
Cloudflare or AI at line 222	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: `codex/fix-ai-worker-cloudflare-fallback` Lets the website AI answer through Cloudflare Workers AI when no OpenAI Worker secret is configured. `development` `development` Active
Installer at line 223	Windows setup, update, shortcut, or installation behavior. Evidence: `codex/bump-builder-version-0.2.26` Bumps the builder release version so the next approved `development` to `main` release creates a fresh installer tag. `development` `development` Active
Compiler or simulator at line 224	Part of the Compile Code workspace or language/tool detection. Evidence: `codex/speed-up-compile-runner` Speeds compiler detection, closes stdin cleanly, reports elapsed compile/run times, shows generated binaries/classes, and wires Icarus Verilog for SystemVerilog simulation. `development`
Compiler or simulator at line 225	Part of the Compile Code workspace or language/tool detection. Evidence: `codex/fix-node-compiler-detection` Prevents the packaged Electron executable from being mistaken for Node.js in the Compile Code workspace. `development` `development` Active
Compiler or simulator at line 226	Part of the Compile Code workspace or language/tool detection. Evidence: `codex/optimize-code-run-cache` Adds cached compiled artifacts and a separate Rebuild / run command so repeated Java, C, C++, and SystemVerilog runs do not recompile unchanged source. `development` `development`
Compiler or simulator at line 227	Part of the Compile Code workspace or language/tool detection. Evidence: `codex/add-verilog-simulator` Adds Verilog as a first-class compile/simulate language using Icarus Verilog and vvp, with Verilog-specific highlighting, file detection, and cleaner simulator output. `development`
Installer at line 231	Windows setup, update, shortcut, or installation behavior. Evidence: `codex/expand-complete-guide-detail` Expands the complete guide with deeper flow walkthroughs, API endpoint details, data contracts, caching detail, installer internals, generated function inventory, and additional b
Security or auth at line 232	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: `codex/hdl-testbench-scope` Requires HDL simulations to include a testbench, adds HDL design/testbench roles, and renders VCD waveform data in a signal scope. `development` `development` Merged into `developmen
Security or auth at line 233	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: `codex/builder-file-ai-auth-docs-cleanup` Allows all project evidence file types, cleans publishing target authentication/loading, improves the website AI fallback, adds IDE-style compile workspace controls, and expa
Installer at line 236	Windows setup, update, shortcut, or installation behavior. Evidence: ## Uninstall
Git command at line 238	Repository state, publishing, branch checks, or release automation. Evidence: Uninstall from Windows **Installed apps** / **Apps & features***. The uninstaller removes OMB Portfolio Builder, its shortcuts, and local app data when Windows requests app-data cleanup. It does not silently remove shared

Representative opening snippet

```

1: # OMB Portfolio Builder
2:
3: OMB Portfolio Builder is a Windows desktop application for creating, editing, previewing, and publishing
a portfolio site from a local workspace. A fresh install starts blank: users add their own profile, sections,
projects, files, images, links, resumes, appearance choices, and publishing target.
4:
5: This README is for the builder application only. Website-specific content, domain notes, public page
layout, and generated portfolio assets belong in the portfolio site's own README.
6:
7: ## Complete Beginner Guide
8:
9: A long beginner-oriented Word guide and a deeper source-code reference are included at:
10:
11: ```text
12: docs/OMB_Portfolio_Builder_Complete_Guide.docx
13: docs/OMB_Portfolio_Builder_Code_Reference.docx
14: docs/OMB_Portfolio_Builder_Code_Reference.pdf
15: ```
16:
17: It explains the high-level design, generated block diagrams, software-engineering decisions, shell
commands, programming syntax, how the important files communicate, what each build tool owns, GitHub workflows,
Electron, the installer, caching, API endpoints, data contracts, frontend/backend/AI/Cloudflare layers, the
website, the parser, publishing, compiler tools, generated function inventories, and file-by-file repository
notes. The generator also writes source-focused Word documents to `docs/code-reference-docx` and PDF documents
to `docs/code-reference-pdf`.
18:
19: ## Install
20:
21: Download the latest Windows installer or portable executable from:
22:
23: ```text
24: https://github.com/otienomaurice/omb-portfolio-builder/releases/latest
25: ```
26:
27: Use:
28:
29: - `OMB-Portfolio-Builder-Setup-latest-x64.exe` for the Windows installer.
30: - `OMB-Portfolio-Builder-Portable-latest-x64.exe` when you want to run without installing.
31:

```

32: The normal installer default is a per-user AppData location:
33:
34: ```text
35: C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder
36: ```
37:
38: The installed application executable is:
39:
40: ```text
41: C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder\OMB Portfolio Builder.exe
42: ```
43:
44: The installer uses a Windows wizard with Back, Next, Cancel, progress, and Finish. It can create Desktop and Start Menu shortcuts, and it writes a normal Windows uninstall entry. The default per-user install does not write into 'C:\Program Files', so normal app updates can run without a Windows administrator prompt.
45:
46: Installed users do not need Node.js or pnpm. The packaged desktop app includes its runtime. Git for Windows with Git Credential Manager is needed only for publishing to a Git repository. If Git or Git Credential Manager is missing, setup can install Git for Windows; compatible existing tools are skipped rather than removed.
47:
48: ## Updates
49:
50: The app checks GitHub Releases for newer builder versions. When a newer release exists, the app shows an update dialog. Choosing **Update** downloads the installer, closes the running builder, updates the existing installation in place, and reopens the updated app.
51:
52: Setup treats an existing install as an update target, not a second installation. It checks the current user's install first, then legacy machine-wide records, uses the detected install location for updates, avoids creating duplicate builder copies, and stops when the installed version is already current or newer.
53:
54: Older releases remain available at:
55:
56: ```text
57: https://github.com/otienomaurice/omb-portfolio-builder/releases
58: ```
59:
60: ## Local Workspaces
61:
62: On Windows, the app keeps application files and editable content under AppData:
63:
64: - `C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder` stores the installed desktop application.
65: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\builder` stores the builder workspace, local drafts, uploaded files, templates, previews, and editing tools.
66: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\portfolio` stores the publish mirror used by Git-based publishing.
67: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\compile-code` stores project compile-workspace source files and run folders created by the Compile Code tool.
68:
69: The installer and updater treat the AppData builder and portfolio folders as managed workspaces, not duplicate app installs.
70:
71: The builder works offline for editing, project creation, file attachment, local previews, and draft saves. Publishing requires network access and repository write permission.
72: